



The Grue Divide: When Cognitive Accuracy Doesn't Predict Affective Alignment

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Overall, Korean speakers counted blue-gradient blocks more accurately than English-speaking cohorts. Yet seeing a color with the same precision doesn't mean feeling it the same way.

1 Does language shape how we see?

The **Sapir-Whorf hypothesis** says the language we speak influences how we think. Its weak form has solid experimental support, and color is its classic testing ground.

	Strong version Linguistic determinism	Weak version Linguistic relativity
Core idea	Language determines thought. Linguistic categories strictly limit what a person can perceive or understand.	Language influences thought. The structure of a mother tongue creates cognitive habits, nudging attention toward certain aspects of reality.
Scientific consensus	Largely rejected. People can grasp concepts even when their language has no word for them.	Widely accepted in a moderate form. Language affects the speed of perception, memory and categorization.

this study Color perception Russian has two basic words for blue, and Russian speakers tell blues apart faster across that line (Winawer et al., 2007).	Time and space Speakers of Guugu Yimithirr (Australia) say “north” and “south” instead of “left” and “right”, and keep track of compass directions all the time (Levinson).	Grammatical gender Does a noun's grammatical gender nudge the adjectives people pick for it? (Boroditsky et al.; findings mixed)
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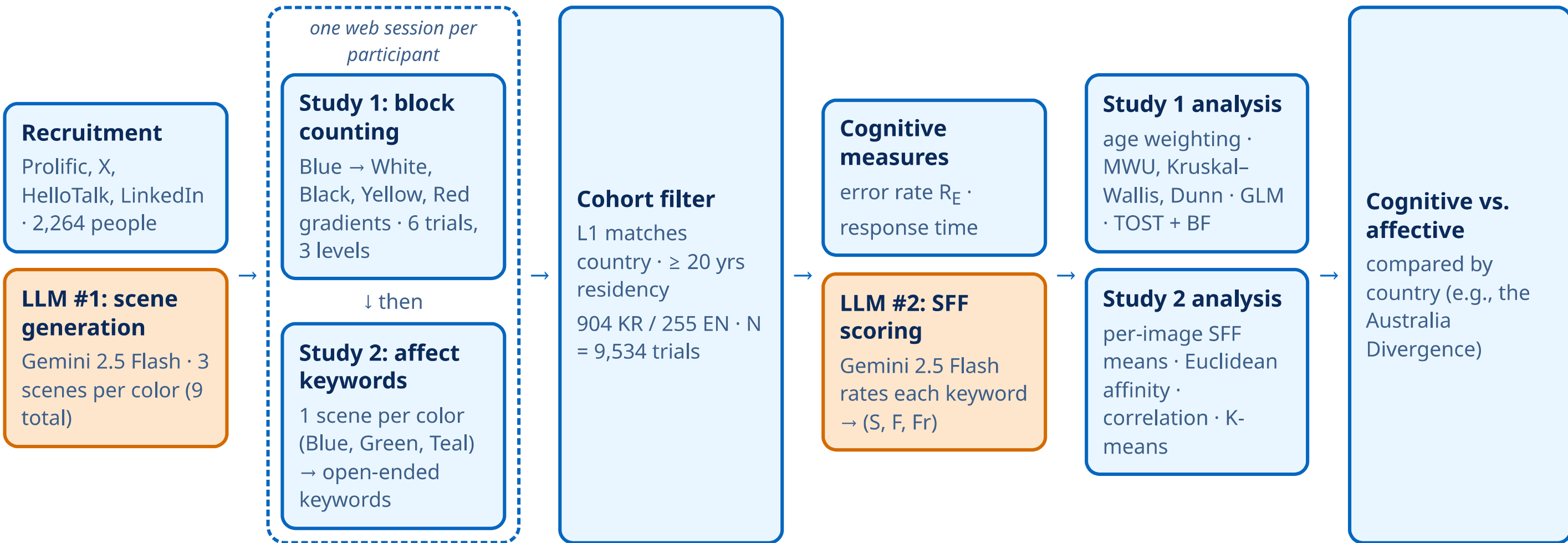
2 Why Grue?



One word for the blue sky and the green forest. Does that mean Korean speakers can't tell green from blue? Our data say no.

One word, two colors. Blue is the anchor of *Obangsaek*, and Grue has the fewest derived forms yet is the most-used Korean color category (Lee and Kim, 1996).

3 Method



Orange = where the LLM enters. Adapted from Figure 6 of the paper.

Accuracy is a difficulty-weighted relative error for each block-counting trial:

$$R_E = \frac{|B_a - E_u|}{B_a \times L_d}$$

Example: a level-2 bar has 20 blocks and the participant says 18 → $|20 - 18| \div (20 \times 2) = 0.05$. A perfect answer scores 0.

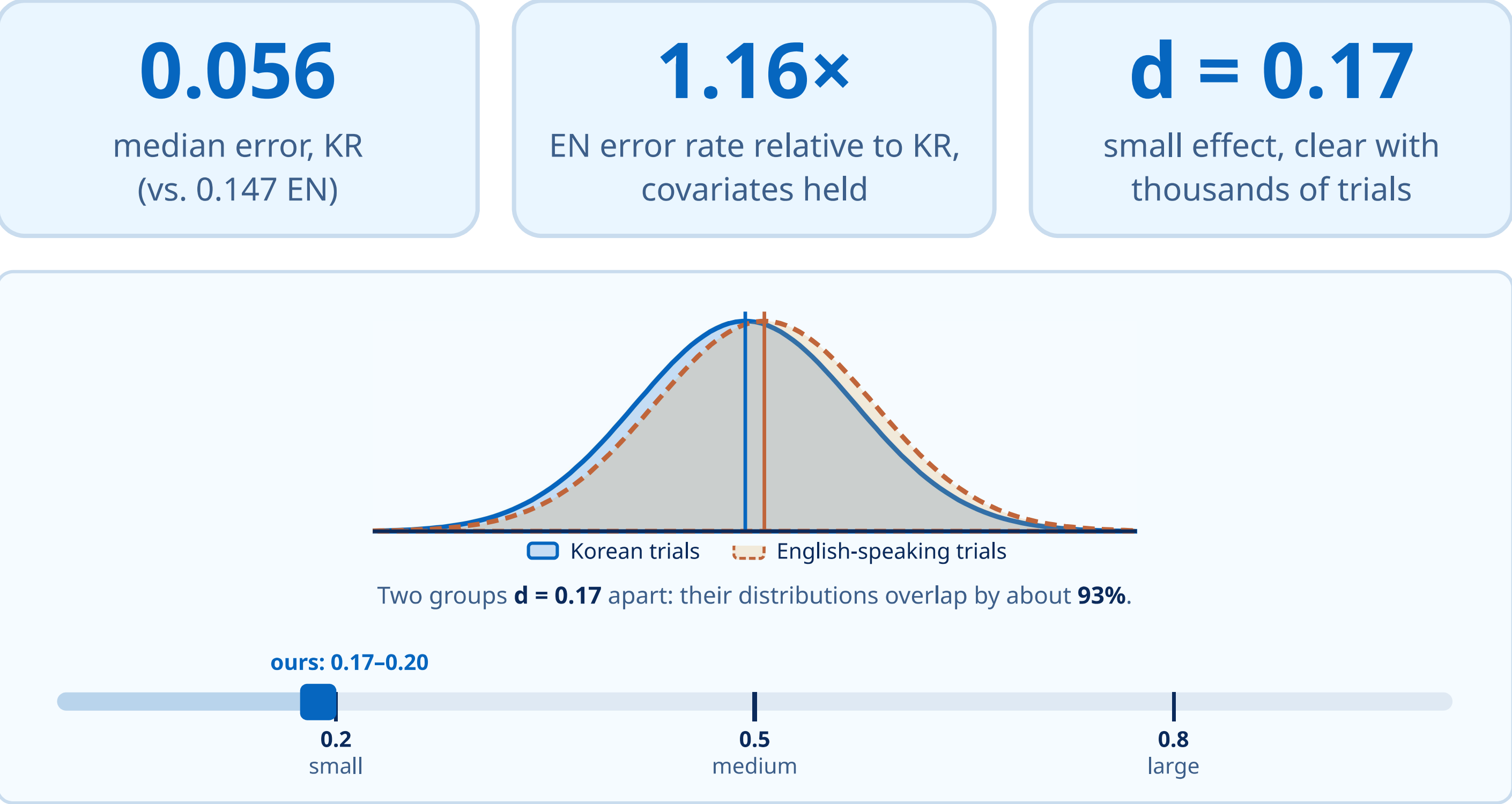
4 Limitations

- Exploratory and correlational; the Australian subsample is small (132 trials).
- SFF scores come from an LLM and aren't yet validated against human raters; Sentiment and Friendliness are highly collinear.

5 The big idea

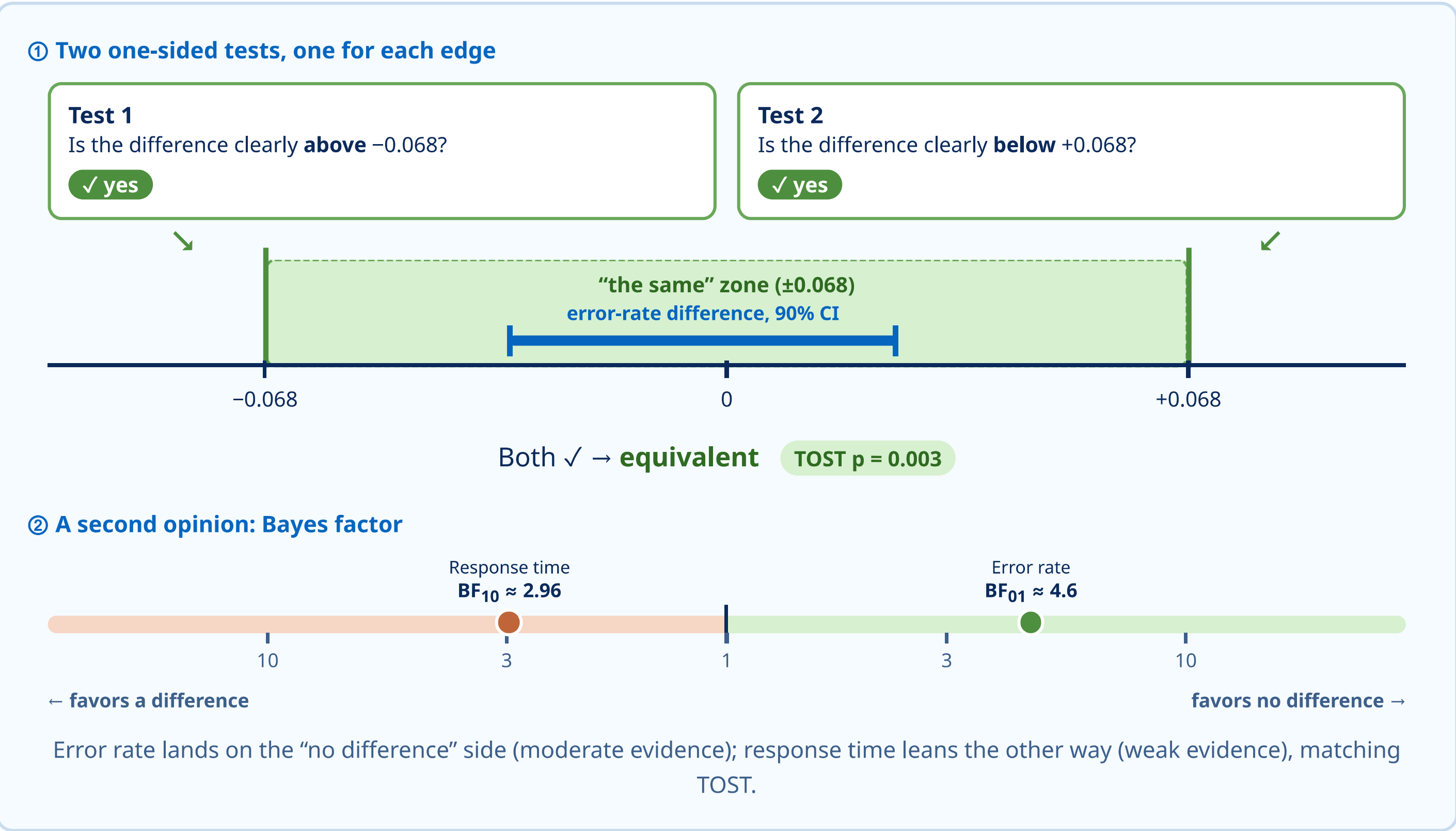
1 The puzzle Korean uses one word, <i>pureuda</i> , for both blue and green. A naive reading of Whorf predicts worse discrimination there. Korean speakers were more accurate.	2 Two clues Speed: faster only in the blue-to-green gradient, where Korean and English boundaries differ. This matches Winawer's Russian finding. Accuracy: better across <i>all four</i> gradients. This fits how productive Korean color words are (many derived forms) and how often Grue is used.	3 So we measure feeling, too Words can't be matched one-to-one across languages (419 Korean variants vs. 20 English derived forms). SFF scores every keyword on the same three axes instead, so affect can be compared directly.
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6 Perception results



7 The Australia Divergence

Korean and Australian participants were **equivalent in accuracy**. In how they described colors emotionally, though, Australians grouped with the US and UK, not with Korea.



8 Measuring feeling: SFF

Gemini 2.5 Flash scored each free-text keyword from -1 to 1 on three axes, stable across temperatures (median cosine ≈ 0.99):

Sentiment valence	Formality social distance	Friendliness warmth
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Take-home: perceptual precision can be shared across cultures while affective mapping stays culturally distinct, so categorical accuracy is not enough for cultural alignment.



Take the test



Read the paper



This website

Data: osf.io/xub35 · **Code:** github.com/bakeyeon/colorfish_public

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